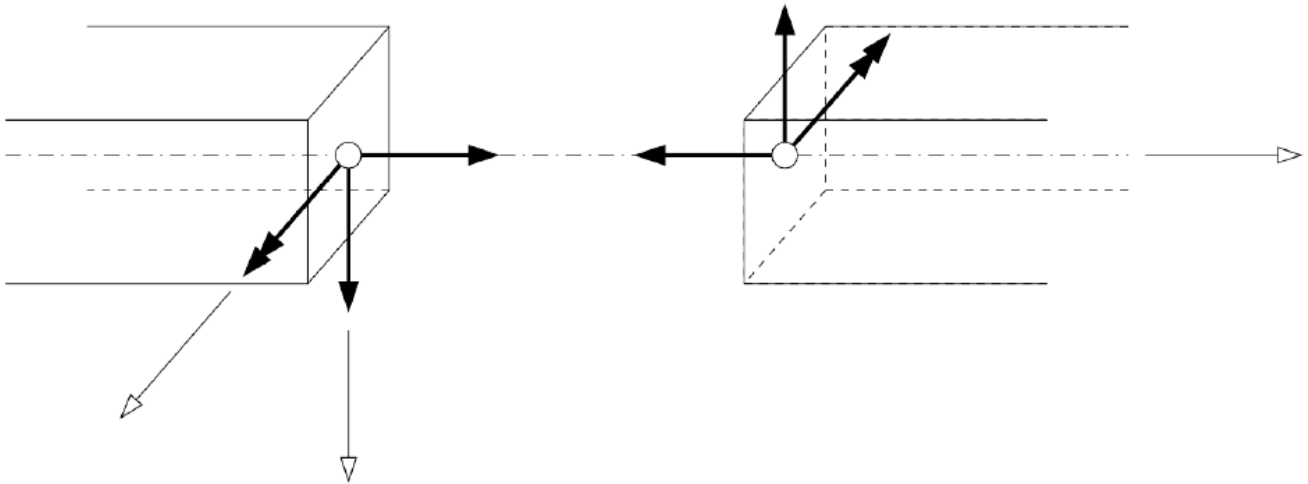
Engineering Mechanics I

Lecture



5.4. Static Determinacy

A System is **statically determined**, if

- all support/bearing reactions can be calculated from the equilibrium conditions.
- all degrees of freedom of the system are uniquely constrained.
- no pole exists for any structural element

➢ two necessary criteria: Counting criterion + Pole criterion

System is **statically indeterminate** if

- more support and intermediate conditions are present than equilibrium conditions

5.4.1. Counting criterion

Box 5.4

S = Number Support Reactions

I = Number Intermediate Reactions

E = Number of elements (in 2D: **Disks**)

- Counting criterion is **necessary condition**
- Another necessary condition: **kinematic determinacy** (System not movable)

⟶ Static determinacy only by **combination of both** conditions

5.4.1. Counting criterion

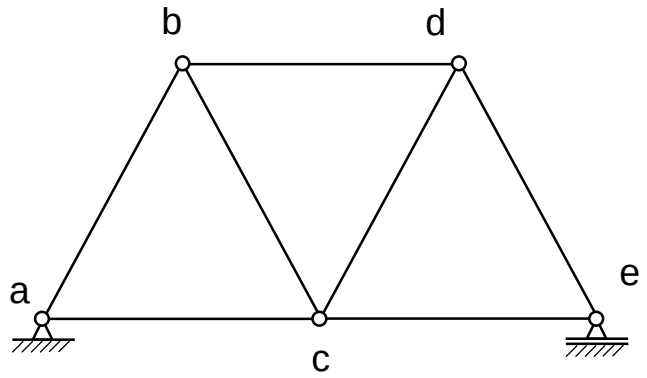
- Number of support reactions, corresponds to the value of the support.
- Value of a joint causes intermediate reactions
- For a moment/ball joint in 2 and 3 dimensions, respectively:

Box 5.5

5.4.1. Counting criterion

Example 5.1: Counting criterion

Sketch:



Counting criterion:

Box 5.6

5.4.2. Pole plans and pole criterion (Instant center of rotation)

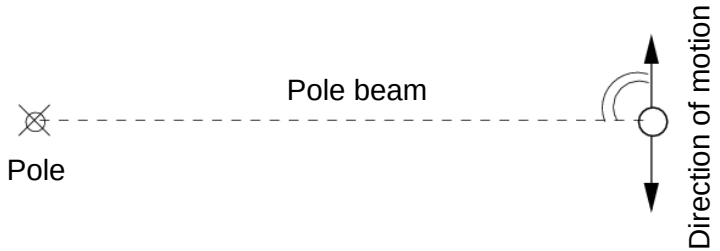
- Pole plans are criterion for displaceability and thus for the 2nd necessary condition for static determinacy of a system.
- If no rotation pole can be found for a rigid body, it is not displaceable and thus at least kinematically determined.

Remarks:

- Importance of poles increases for dynamic systems.
- So-called *instantaneous* poles are added.

5.4.2. Pole plans and pole criterion

- Infinitesimal displacement is always also rotation around (rotational) pole
- Translational displacement has pole at infinity
- With knowledge of the rotation pole kinematics of a system can be described



- Displacement always perpendicular to *pole beam*

  à defines geometric location for pole

5.4.2. Pole plans and pole criterion

- With two known displacement directions, the associated pole is located at the intersection of the two pole rays
- This allows to create a **pole plan** for the whole system
- Definition:

  - **Main pole**: Pole of rotation of a rigid body around which it rotates. The pole itself does not move. Designation (0,i)

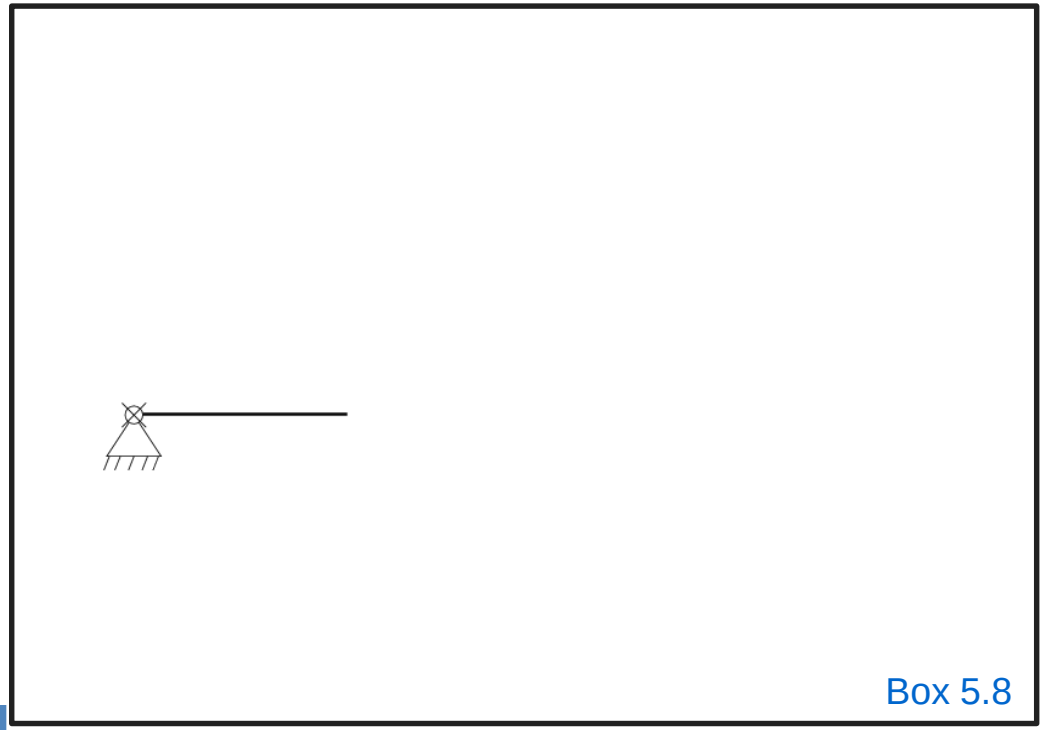
  - **Relative pole** (minor pole): Pole around which a rigid body i rotates with respect to another rigid body j. Designation (i,j)
- The two main poles and the common secondary pole lie on a straight line

5.4.2. Pole plans and pole criterion

- With two known displacement directions, the associated pole is located at the intersection of the two pole rays
- This allows to create a **pole plan** for the whole system
-
-

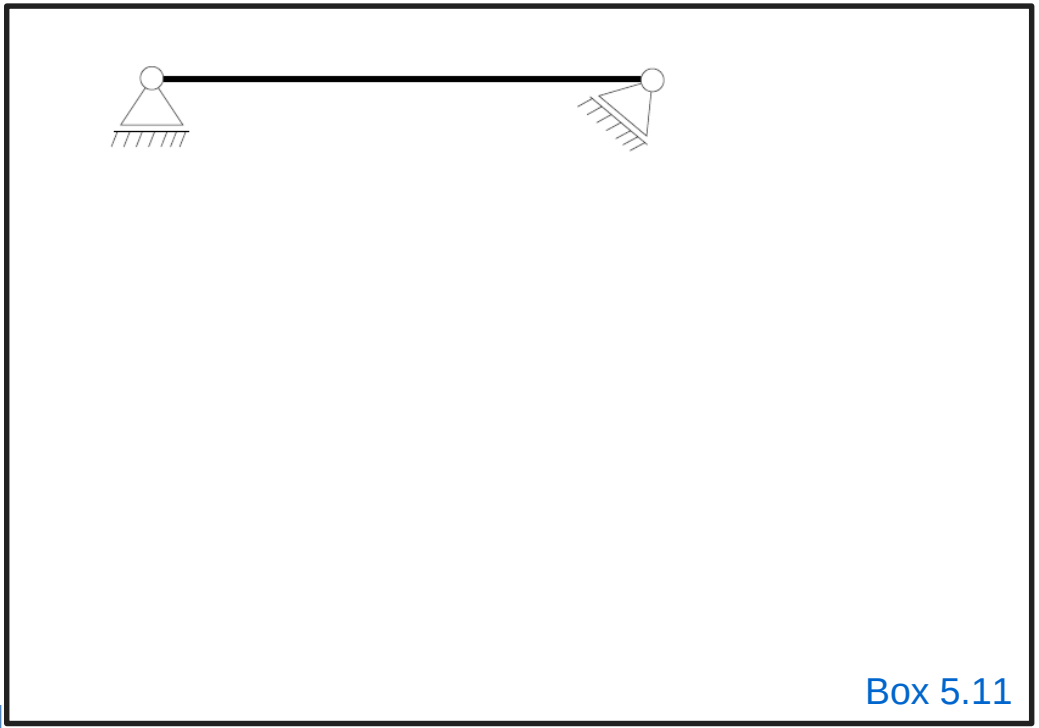
Box 5.7

5.4.2. Pole plans and pole criterion - Examples



- Indisplaceable rotary supports are always poles of the connected disks.

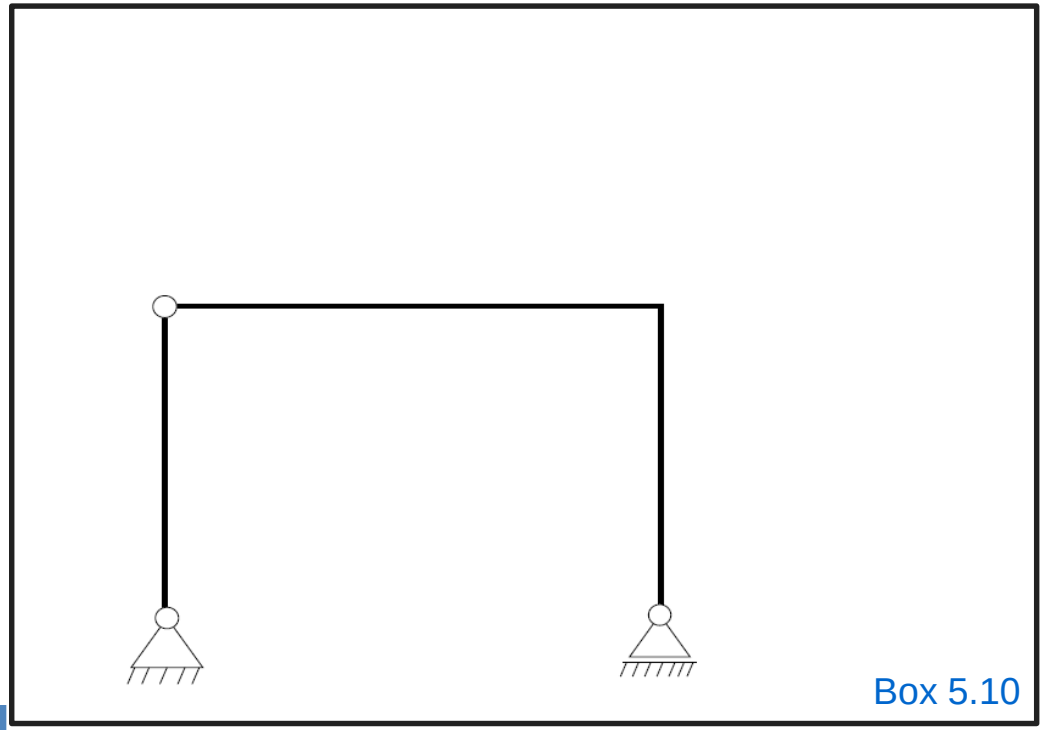
5.4.2. Pole plans and pole criterion - Examples



Box 5.11

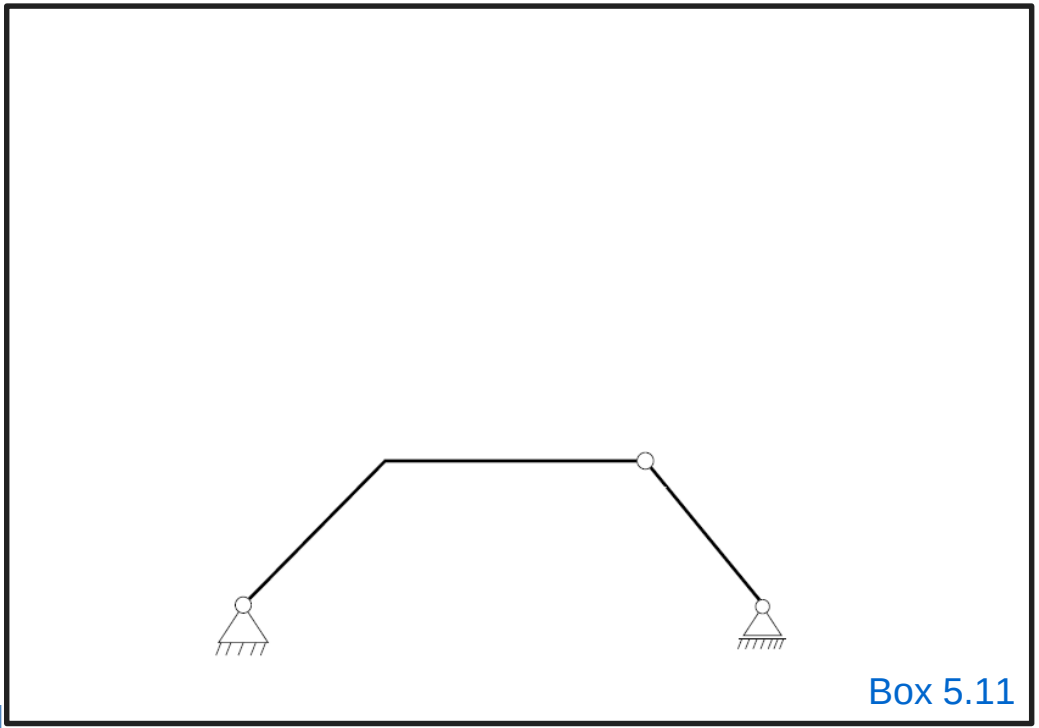
- The perpendicular to a possible direction of displacement of a disk, e.g. by a displaceable support, is a geometric location for the sought pole of rotation.

5.4.2. Pole plans and pole criterion - Examples



- If the pole of a disk lies in infinity (far point), the disk can only experience a translation, but no rotation.
- Parallel lines intersect at infinity.

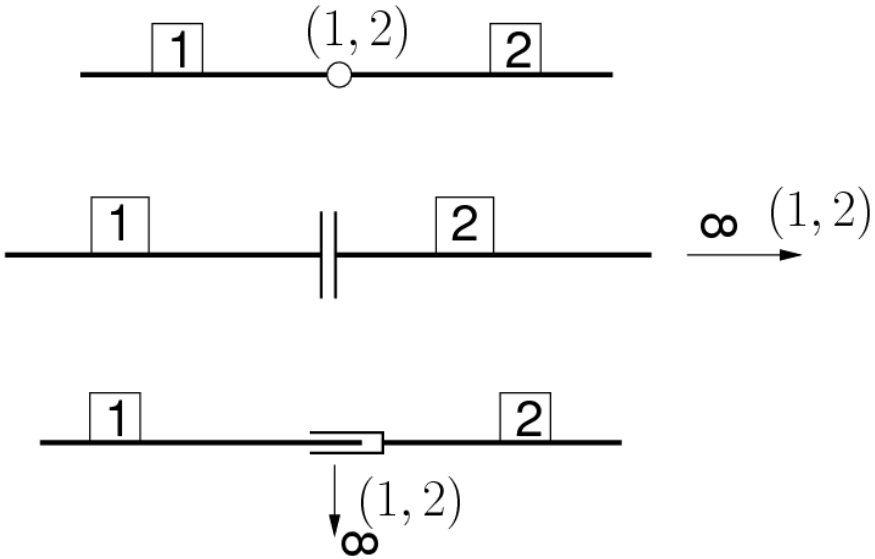
5.4.2. Pole plans and pole criterion - Examples



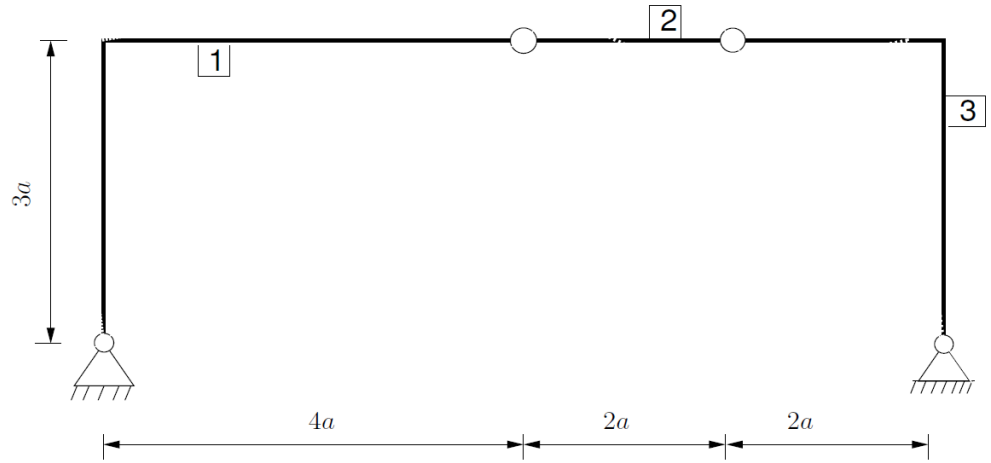
- The pole of a disk, the connection point (minor pole) to a second disk (joint) and the pole of this second disk lie on a straight line.

5.4.2. Pole plans and pole criterion - Examples

Joints, minor poles



5.4.2. Pole plans and pole criterion – Example Full System



Box 5.12

5.4.2. Pole plans and pole criterion – Example Full System

Pole Criterion

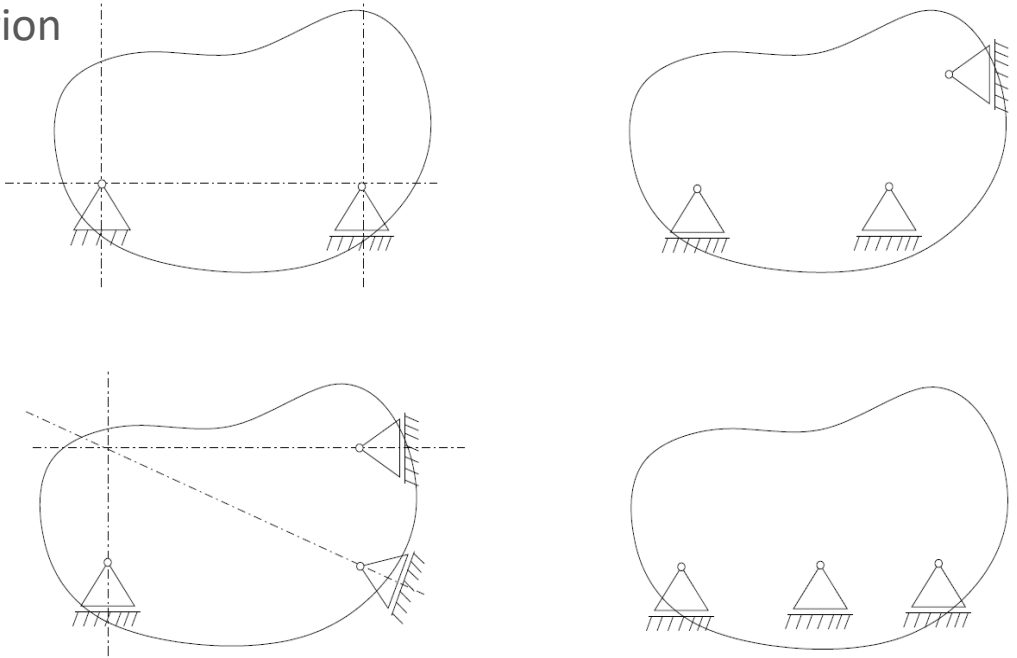
- Pole diagram describes displaceability of the system

- Criterion for displaceability derivable
  - No pole for a body à Body is immovable.
  - System is unmovable if all parts of the system are unmovable.

- Pole Criterion: "*A structure is kinematically determinate if it is non-displacable, i.e., if the pole plan for the structure cannot be constructed without contradiction.*"

5.4.2. Pole plans and pole criterion

Example 5.2: Pole Criterion

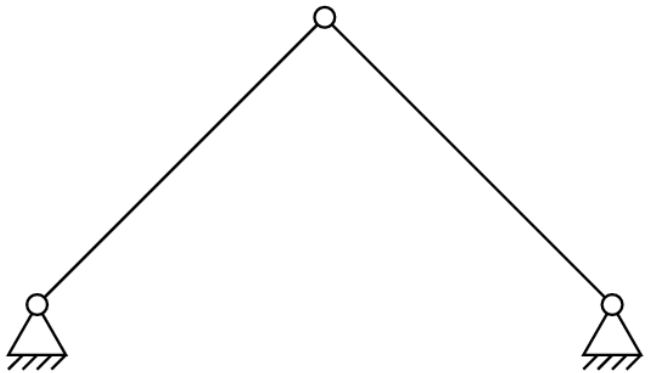


Additionally, counting criterion as a necessary, not sufficient condition.

5.4.2. Pole plans and pole criterion

Example 5.3: Static determinacy of a bar

Pole Plan:



Counting Criterion

Box 5.13

à Counting criterion **and** pole criterion fulfilledà statically determined

5.4.2. Pole plans and pole criterion

Summary on static determinacy

- Statically determinate only if counting criterion and pole criterion are fulfilled.

- Statically indeterminate, if more support and intermediate reactions are present than can be calculated with the equilibrium relations.